SI 140A.02 Probability & Statistics for EECS, Fall 2026
Homework 5

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to submit your homework in PDF format generated by **LaTeX**, otherwise you will get no points of this homework.
- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1 (20 points)

A/B testing is a randomized experiment used to compare two treatments, Treatment A and Treatment B. Users arrive one by one and are assigned to one of the treatments. Each trial is classified as either a success or a failure.
Consider the following "stay-with-a-winner" procedure:

(i) Assign the first user to Treatment A or Treatment B with equal probabilities.

(ii) If the trial for the nth user is a success, assign the $(n+1)$st user to the same treatment. Otherwise, switch to the other treatment.

The success probabilities for Treatments A and B are a and b, respectively, where $a \neq b$. Given the assigned treatment, these success probabilities do not depend on the previous outcomes.
Let p_n be the probability of success on the nth trial, and let a_n be the probability that Treatment A is assigned on the nth trial.

(a) Show that
$$p_n = (a-b)a_n + b, \qquad a_{n+1} = (a+b-1)a_n + 1 - b.$$

(b) Use part (a) to show that
$$p_{n+1} = (a+b-1)p_n + a + b - 2ab.$$

(c) Assuming that the limit exists, find
$$\lim_{n\to\infty} p_n.$$

Solution

Part A

Conditioning on the treatment assigned on trial n,

$$p_n = a\,a_n + b(1-a_n) = (a-b)a_n + b.$$

For a_{n+1}, note that Treatment A is used on trial $n+1$ if either A was used on trial n and succeeded, or B was used on trial n and failed. Hence

$$a_{n+1} = a\,a_n + (1-b)(1-a_n) = (a+b-1)a_n + 1 - b.$$

Part B

Using part (a) twice,

$$\begin{aligned} p_{n+1} &= (a-b)a_{n+1} + b = (a-b)\big[(a+b-1)a_n + 1 - b\big] + b \\ &= (a+b-1)(p_n - b) + (a-b)(1-b) + b \\ &= (a+b-1)p_n - b(a+b-1) + (a-b)(1-b) + b \\ &= (a+b-1)p_n + a + b - 2ab. \end{aligned}$$

Part C

If $p_n \to p$, taking limits in part (b) gives

$$p = (a+b-1)p + a + b - 2ab \implies (2-a-b)p = a + b - 2ab,$$

and therefore

$$\lim_{n\to\infty} p_n = \frac{a+b-2ab}{2-a-b}.$$

Problem 2 (20 points)

A fair coin is tossed repeatedly and independently, producing a sequence of heads and tails. Let N be the number of tosses required to observe the first occurrence of the pattern HH.
Find the PMF of N.

Solution

Let $f_n = \Pr(N = n)$. Clearly $f_1 = 0$ and $f_2 = \Pr(HH) = 1/4$.
For $n \geq 3$, condition on the first two tosses:

- if the first toss is T (probability $1/2$), the process restarts, contributing $\frac{1}{2} f_{n-1}$;
- if the first two tosses are HT (probability $1/4$), the process restarts after two tosses, contributing $\frac{1}{4} f_{n-2}$.

Hence

$$f_n = \frac{1}{2} f_{n-1} + \frac{1}{4} f_{n-2}, \qquad n \geq 3.$$

The characteristic equation $r^2 = \frac{1}{2} r + \frac{1}{4}$ has roots

$$r = \frac{1 \pm \sqrt{5}}{4}.$$

Writing $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$, the general solution is

$$f_n = A \left(\frac{\varphi}{2} \right)^n + B \left(\frac{\psi}{2} \right)^n.$$

The initial conditions $f_1 = 0$ and $f_2 = 1/4$ give

$$A\varphi + B\psi = 0, \qquad A\varphi^2 + B\psi^2 = 1,$$

whose solution is

$$A = \frac{1}{\sqrt{5}\,\varphi}, \qquad B = -\frac{1}{\sqrt{5}\,\psi}.$$

Therefore

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{\varphi^{n-1}}{2^n} - \frac{\psi^{n-1}}{2^n} \right) = \frac{\varphi^{n-1} - \psi^{n-1}}{\sqrt{5}\,2^n} = \frac{F_{n-1}}{2^n},$$

where

$$F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}$$

is the kth Fibonacci number ($F_0 = 0$, $F_1 = F_2 = 1$). Hence

$$\Pr(N = n) = \frac{F_{n-1}}{2^n}, \qquad n \geq 1.$$

Problem 3 (15 points)

A system contains seven devices: four in Group A and three in Group B. Each device fails independently with probability p, where $0 < p < 1$.
Let X be the number of failed devices in Group A, and let T be the total number of failed devices in the system.

(a) Find the PMF of X and the PMF of T.

(b) Given that exactly three devices have failed, find the conditional PMF of X. Identify the distribution.

(c) Explain why the conditional distribution in part (b) does not depend on p.

Solution

Part A

Each device fails independently with probability p, so

$$\Pr(X = i) = \binom{4}{i} p^i (1-p)^{4-i}, \qquad i = 0, 1, 2, 3, 4,$$

and

$$\Pr(T = j) = \binom{7}{j} p^j (1-p)^{7-j}, \qquad j = 0, 1, \ldots, 7.$$

That is, $X \sim \text{Binomial}(4, p)$ and $T \sim \text{Binomial}(7, p)$.

Part B

For $i = 0, 1, 2, 3$,

$$\begin{aligned}
\Pr(X = i \mid T = 3) &= \frac{\Pr(T = 3 \mid X = i)\Pr(X = i)}{\Pr(T = 3)} \\
&= \frac{\binom{3}{3-i} p^{3-i}(1-p)^i \binom{4}{i} p^i (1-p)^{4-i}}{\binom{7}{3} p^3 (1-p)^4} \\
&= \frac{\binom{4}{i}\binom{3}{3-i}}{\binom{7}{3}},
\end{aligned}$$

where $\Pr(T = 3 \mid X = i)$ corresponds to exactly $3 - i$ failures among the three devices of Group B. Numerically,

$$\Pr(X = i \mid T = 3) = \frac{1}{35}, \ \frac{12}{35}, \ \frac{18}{35}, \ \frac{4}{35} \qquad (i = 0, 1, 2, 3).$$

This is the hypergeometric distribution: given that three of the seven devices failed, the failing devices form a uniformly random three-element subset of the seven, and X counts how many of them belong to the four devices of Group A.

Part C

Since all seven devices have the same failure probability p and fail independently, conditioning on the total number of failures makes every subset of that size equally likely; the value of p affects only how many devices fail, not which ones fail. Thus the conditional PMF in part (b) does not involve p.

Problem 4 (15 points)

Two fair six-sided dice are rolled independently. Let U and V be their respective outcomes, and define

$$D = |U - V|.$$

(a) Find the PMF of D.

(b) Find the CDF $F_D(t)$ for every real number t.

(c) Define $W = \min(D, 2)$. Find the PMF of W.

Solution

Part A

There are 36 equally likely outcomes (U, V). For $i = 1, \ldots, 5$ there are $2(6 - i)$ outcomes with $|U - V| = i$, while $D = 0$ occurs for the 6 outcomes with $U = V$. Hence

$$\Pr(D = i) = \begin{cases} \dfrac{1}{6}, & i = 0, \\ \dfrac{6 - i}{18}, & i = 1, 2, 3, 4, 5, \\ 0, & \text{otherwise.} \end{cases}$$

Part B

Accumulating the PMF from part (a),

$$F_D(t) = \begin{cases} 0, & t < 0, \\ \dfrac{1}{6}, & 0 \le t < 1, \\ \dfrac{4}{9}, & 1 \le t < 2, \\ \dfrac{2}{3}, & 2 \le t < 3, \\ \dfrac{5}{6}, & 3 \le t < 4, \\ \dfrac{17}{18}, & 4 \le t < 5, \\ 1, & t \ge 5. \end{cases}$$

Part C

Since $W = \min(D, 2)$,

$$\Pr(W = 0) = \Pr(D = 0) = \frac{1}{6}, \qquad \Pr(W = 1) = \Pr(D = 1) = \frac{5}{18},$$

and

$$\Pr(W = 2) = \Pr(D \ge 2) = 1 - \frac{1}{6} - \frac{5}{18} = \frac{5}{9}.$$

Thus $\Pr(W = w) = 0$ for $w \notin \{0, 1, 2\}$.

Problem 5 (15 points)

There are two coins, A and B. Coin A lands heads with probability 1/4, while coin B lands heads with probability 3/4.
One coin is selected with equal probabilities and then tossed twice. The same selected coin is used for both tosses. Given the selected coin, the two tosses are independent.
Let $C \in \{A, B\}$ denote the selected coin. Let X and Y be the indicator random variables for heads on the first and second tosses, respectively.

(a) Find the PMF of X and the PMF of Y.

(b) Are X and Y independent? Are they conditionally independent given C? Justify your answers.

(c) Given that both tosses are heads, find the conditional probability that coin B was selected.

Solution

Part A

By the law of total probability,

$$\Pr(X = 1) = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{2},$$

so $\Pr(X = 0) = \frac{1}{2}$. The two tosses use the same coin and therefore have the same marginal distribution, so Y has the same PMF:

$$\Pr(Y = 1) = \Pr(Y = 0) = \frac{1}{2}.$$

Part B

The variables X and Y are not independent. Indeed,

$$\Pr(X = 1, Y = 1) = \frac{1}{2}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2 = \frac{1}{32} + \frac{9}{32} = \frac{5}{16},$$

whereas

$$\Pr(X = 1)\Pr(Y = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \neq \frac{5}{16}.$$

They are conditionally independent given C, because given the selected coin the two tosses are independent:

$$\Pr(X = x, Y = y \mid C) = \Pr(X = x \mid C)\Pr(Y = y \mid C)$$

for all $x, y \in \{0, 1\}$.

Part C

By Bayes' rule,

$$\begin{aligned}\Pr(C = B \mid X = 1, Y = 1) &= \frac{\Pr(X = 1, Y = 1 \mid C = B)\Pr(C = B)}{\Pr(X = 1, Y = 1)} \\ &= \frac{\frac{9}{16} \cdot \frac{1}{2}}{\frac{5}{16}} = \frac{9}{10}.\end{aligned}$$

Problem 6 (15 points)

A random variable X has PMF

$$\Pr(X=1)=\frac{1}{2},\qquad \Pr(X=2)=\frac{1}{4},\qquad \Pr(X=3)=\Pr(X=4)=\frac{1}{8}.$$

Let U have the discrete uniform distribution on $\{1,2,3,4\}$.
Throughout this problem, logarithms have base 2, and entropy is measured in bits.

(a) Find $H(X)$ and $H(U)$.

(b) Define

$$Y=\begin{cases}1, & X\le 2,\\ 0, & X>2,\end{cases}\qquad Z=\begin{cases}1, & X=1,\\ 0, & X\neq 1.\end{cases}$$

Find the PMF of each variable and calculate $H(Y)$ and $H(Z)$.

(c) Is the size of the support sufficient to determine the entropy of a discrete random variable? Explain using distributions from this problem.

Solution

Part A

$$H(X)=-\sum_{i=1}^{4}\Pr(X=i)\log_2\Pr(X=i)=\frac{1}{2}\log_2 2+\frac{1}{4}\log_2 4+2\cdot\frac{1}{8}\log_2 8=\frac{1}{2}+\frac{1}{2}+\frac{3}{4}=\frac{7}{4}=1.75\text{ bits},$$

and since U is uniform on a set of size 4,

$$H(U)=\log_2 4=2\text{ bits}.$$

Part B

Since $\Pr(Y=1)=\Pr(X\le 2)=\frac{1}{2}+\frac{1}{4}=\frac{3}{4}$ and $\Pr(Y=0)=\frac{1}{4}$,

$$H(Y)=-\frac{3}{4}\log_2\frac{3}{4}-\frac{1}{4}\log_2\frac{1}{4}=\frac{3}{4}\log_2\frac{4}{3}+\frac{1}{2}=2-\frac{3}{4}\log_2 3\approx 0.811\text{ bits}.$$

Since $\Pr(Z=1)=\Pr(X=1)=\frac{1}{2}$ and $\Pr(Z=0)=\frac{1}{2}$,

$$H(Z)=-\frac{1}{2}\log_2\frac{1}{2}-\frac{1}{2}\log_2\frac{1}{2}=1\text{ bit}.$$

Part C

No. The support size only gives the upper bound $H\le\log_2|\text{support}|$, with equality exactly for the uniform distribution on the support; it does not determine the entropy. In this problem X and U both have support $\{1,2,3,4\}$, but $H(X)=1.75<2=H(U)$. Similarly, Y and Z both have a support of size 2, but $H(Y)\approx 0.811<1=H(Z)$.